Mathematics for Game Theory

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1. Continuity of real functions

You might recall that a real function is continuous if you can draw it without lifting the pen. That gives a nice intuition but we will have to go a bit deeper. Roughly speaking, a real function f is continuous at a point x if f maps points close to x to values close to $f(x)$. More formally, f is continuous at x if for every $\varepsilon > 0$ there exists a $\delta > 0$ (which might depend on x and ε) such that $|f(x') - f(x)| < \varepsilon$ for all x' such that $|x' - x| < \delta$. Put differently, for every ε ball around $f(x)$ we can find a δ ball around x such that all points in this δ ball have images in the ε ball. If a function is continuous at all x in its domain, we say *the function is continuous.*

Let's show that the linear function $f(x) = 2x$ is continuous. Take any $\varepsilon > 0$ and some x. I propose to use $\delta = \varepsilon/2$. Then, $|f(x) - f(x')| = |2x - 2x'| = 2|x - x'| < 2\delta = \varepsilon$ which is what we had to show.

Let's show the same for the function $g(x) = x^2$. Let us take $x = 2$ and $\varepsilon > 0$ (and let us immediately focus on $\varepsilon < 1$ as smaller ε make it harder to show continuity, right?). I propose $\delta = \varepsilon/8$. Then $|g(x) - g(x')| = |2^2 - x'^2| < |2^2 - (2 + \delta)^2| = 4\delta + \delta^2 = 4\varepsilon/8 + \varepsilon^2/64 = \varepsilon/2 + \varepsilon^2/64 < \varepsilon$ (as $\varepsilon < 1$). Exercise for you: Show that g is also continuous at $x = 5$ (it will hopefully not surprise you to hear the g is in fact continuous at all x).

I want to point out one result for composite functions: Say f and g are continuous functions. Then the composite function h defined by $h(x) = f(g(x))$ is also continuous. For example, e^x is continuous and $sin(x)$ is continuous, and therefore $e^{sin(x)}$ is also continuous. The proof of the general result is not very difficult: Take an arbitrary x

and $\varepsilon > 0$. Since f is continuous, we can find a $\delta' > 0$ such that all y in the δ' ball around $g(x)$ map into the ε ball around $f(g(x))$. Now since g is continuous we can find a $\delta > 0$ such that all x' in the δ ball around x map into the δ' ball around $g(x)$. Taking these two things together, all x' in the δ ball around x map into the ε ball around $h(x)$ which proves that h is continuous.

Another important result is the following: Say f and g are continuous functions. Then the function h defined by $h(x) = f(x) + g(x)$ is also continuous. The proof is again not too hard: Take an arbitrary x and $\varepsilon > 0$. As f is continuous, we can find a $\delta' > 0$ such that all x' in the δ' ball around x map into the $\varepsilon/2$ ball around $f(x)$ under f. Similarly, we can find a $\delta'' > 0$ such that all x' in the δ'' ball around x map into the $\varepsilon/2$ ball around $g(x)$ under g. Taking $\delta = \min\{\delta', \delta''\}$, it follows that all x' in the δ ball around x map into the ε ball around $h(x)$ under h.

From these results, several other results can be derived. We will use the following:

- All polynomial functions (of one or several variables) are continuous. This includes in particular all linear functions.
- If g and f are continuous and $f > 0$, then g/f is continuous.

2. Uniformly distributed continuous random variables

Roughly speaking, a random variable θ is uniformly distributed if every value in its support is equally likely. The *support* of a random variable is the set of values that are possible. Let's concentrate on the case where the support is an interval $[a, b]$. This means that the random variable will not take values below a or above b. "Equally likely" implies that the probability that θ is in $[x, x + \varepsilon)$ does not depend on x but only on ε (as long as the whole interval $[x, x + \varepsilon)$ is in the support).

The distribution of a continuous random variable is described by the *cumulative distribution function* (cdf) F. The real function F gives the probability that the random variable is below θ, i.e. $F(\theta')$ is the probabiltiy that the random variable θ is below the value θ'. If F is differentiable, then its derivative f is called the *density function.*

Using its cdf, we can define a a uniformly distributed random variable more precisely: A random variable θ is uniformly distributed on the support $[a, b]$ if and only if its

distribution function $F(\theta)$ is

$$F(\theta) = \begin{cases} \frac{\theta - a}{b - a} & \text{if } \theta \in [a, b] \\ 0 & \text{if } \theta < a \\ 1 & \text{if } \theta > b. \end{cases}$$

The density of a uniformly distributed random variable is therefore constant on its support, i.e. $f(\theta) = 1/(b - a)$ if $\theta \in [a, b]$ and $f(\theta) = 0$ if $\theta \notin [a, b]$.

An often used special case is the uniform distribution on $[0, 1]$, i.e. $a = 0$ and $b = 1$. In this case,

$$F(\theta) = \begin{cases} \theta & \text{if } \theta \in [0, 1] \\ 0 & \text{if } \theta < 0 \\ 1 & \text{if } \theta > 1 \end{cases}$$

and

$$f(\theta) = \begin{cases} 1 & \text{if } \theta \in [0, 1] \\ 0 & \text{else .} \end{cases}$$

Now I want to remind you how to take expectations of continuous random variables. Let θ be uniformly distributed on $[0, 1]$. Intuitively, you know that the expectation is $1/2$. But how would you calculate this? The expected value can be computed as

$$\int_0^1 \theta * 1 \, d\theta = \left[\frac{\theta^2}{2}\right]_0^1 = \frac{1}{2} - 0 = 1/2.$$

In general, if a random variable θ is distributed with density f on $[\underline{\theta}, \bar{\theta}]$, the expected value of θ is given by

$$\int_{\underline{\theta}}^{\bar{\theta}} \theta f(\theta) \, d\theta.$$

Let's get back to our uniform example. Now suppose you have to pay 1 to participate in a lottery that pays you 3θ. What is your expected payoff of this lottery? Again this is intuitively $3/2 - 1 = 0.5$ but how do we compute it?

$$\int_0^1 (3\theta - 1) * 1 \, d\theta = \left[3\theta^2/2 - \theta\right]_0^1 = 3/2 - 1 = 0.5.$$

What if the lottery pays $2\theta^2$ instead of 3θ? Then the general formula

$$\mathbf{E}[u(\theta)] = \int_{\underline{\theta}}^{\bar{\theta}} u(\theta) f(\theta) \, d\theta$$

helps us (here u is some payoff function that tells us how much money you will have in state θ). In our example, $f(\theta) = 1$ and the payoff is $u(\theta) = 2\theta^2 - 1$ (i.e. in state θ you get $2\theta^2$ but you had to pay 1 to participate). This means your expected payoff from this lottery is

$$\int_0^1 (2\theta^2 - 1) * 1 \, d\theta = \left[2\theta^3/3 - \theta\right]_0^1 = 2/3 - 1 - 0 = -1/3.$$

3. Bayes' rule

Bayes' rule allows you to calculate conditional probabilities and conditional expectations. We consider some examples first.

Example 1: Let's say a friend of yours roles a fair dice (and you cannot see what the result is). He tells you that the number he got is not 6. What is the probability that he got an odd number? The numbers that are still possible are 1, 2, 3, 4, 5. Three of those five numbers are odd and 2 are even. As all numbers are equally likely, the probability of an odd number should be 3/5.

Example 2: A drug test is 90% accurate in the following sense: If someone is a drug user the drug test will be positive in 90% of the cases and negative in 10% of the cases. If someone is not a drug user the drug test will be negative in 90% of the cases and positive in 10% of the cases. 20% of the poulation use drugs. What is the probability that someone testing positive is a drug user? The way to think about this is the following: Imagine we have a population of 100 people, 20 of which use drugs. If we tested all of them, we would expect to test 80*0.1=8 of the 80 non-drug users positive. We would also expect to test 20*0.9=18 of the 20 drug users positive. Hence, we would expect to test 8+18=26 people positive, 18 of which actually use drugs. Hence, the probability that a positively tested person uses drugs is $18/26 \approx 69\%$.

Bayes' rule helps us to these kind of calculations. It states:

$$prob(A|B) = \frac{prob(B|A)\, prob(A)}{prob(B)} = \frac{prob(A \text{ and } B)}{prob(B)}$$

The events A and B in the second example are

- A: an individual uses drugs

- B: an individual tests positive.

To get the probability that someone tested positive uses drugs $prob(A|B)$ we therefore need the probability that someone who uses drugs tests positive ($prob(B|A)$) which is 0.9; we need the probability that someone uses drugs ($prob(A)$) which is 0.2; and the probability that someone tests positive ($prob(B)$) which is 0.8*0.1+0.2*0.9=0.26. Now we get $0.9 * 0.2/0.26 \approx 69\%$, i.e. we calculated correctly above.

Now let us try something with continuous random variables: Say θ is uniformly distributed on $[0, 1]$. Someone observes the drawn θ and tells you that θ is above 0.2. What is the probability that θ is less than 0.5? Here A is the event $\theta < 0.5$ and B is the event $\theta > 0.2$. We need the probability that θ is both above 0.2 and below 0.5 ($prob(A\ and\ B)$) which is $F(0.5) - F(0.2) = 0.3$ and the probability that $\theta > 0.2$, i.e. ($prob(B)$), which is $1 - F(0.2) = 0.8$. Consequently, $prob(A|B) = 0.3/0.8 = 0.375$.

What is the expected value of θ given that $\theta > 0.2$? Intuitively, all values of θ between 0.2 and 1 are equally likely. Hence, the expected value should be the mid-point between 1 and 0.2 which is 0.6. More formally, the conditional distribution of θ (conditional on $\theta > 0.2$) is uniform on $(0.2, 1.0]$. This distribution has a density of $\tilde{f}(\theta) = 1/(1 - 0.2) = 1.25$ on $(0.2, 1.0]$ (and zero everywhere else). The expected value of this distribution is

$$\int_{0.2}^{1.0} \theta \tilde{f}(\theta)\, d\theta = \int_{0.2}^{1.0} \theta 1.25\, d\theta = 1.25(0.5 * 1^2 - 0.5 * 0.2^2) = 1.25 * 0.48 = 0.6.$$

4. Convergence of real sequences

We start by defining what we mean when we say a sequence *converges* (if you have no idea what a "sequence" is, you might want to check the wikipedia entry `https://en.wikipedia.org/wiki/Sequence`; note that in this handout we are interested in "infinite sequences" only): A sequence $(a_n)_{n=1}^{\infty}$, where each $a_n \in \mathbb{R}$, converges to $a \in \mathbb{R}$ if for any $\varepsilon > 0$ we can find a N such that

$$|a_n - a| \leq \varepsilon \quad \text{for all} \quad n \geq N. \tag{1}$$

We then say that "a is the *limit* of the sequence $(a_n)_{n=1}^{\infty}$".

Example 1. *The sequence* $2, \frac{3}{2}, \frac{4}{3}, \frac{5}{4}, \ldots, \frac{n+1}{n}, \ldots$ *converges to 1: For a given* $\varepsilon > 0$, $\left|\frac{n+1}{n} - 1\right|$ *is less than* ε *for* n *high enough. More precisely,* $\left|\frac{n+1}{n} - 1\right| = \left|\frac{1}{n}\right| = \frac{1}{n} < \varepsilon$ *for* $n > \frac{1}{\varepsilon}$. *Hence, by choosing* $N = \left[\frac{1}{\varepsilon}\right] + 1$ *(where* $[\,]$ *indicates rounding to the next integer) we know that* $|a_n - 1| \leq \varepsilon$ *for all* $n \geq N$ *and therefore the limit of the sequence is 1.*

However, the members of a sequence do not have to be real numbers. We can also have sequences of vectors: A sequence $(a_n)_{n=1}^{\infty}$, where each $a_n \in \mathbb{R}^m$, converges to a vector $a \in \mathbb{R}^m$ if for any $\varepsilon > 0$ we can find a N such that

$$||a_n - a|| \leq \varepsilon \quad \text{for all} \quad n \geq N. \tag{2}$$

In the last equation "$||\cdot||$" denotes a norm, e.g. for a vector $b = (b_1, \ldots, b_m)$ the Euclidean norm is

$$||b|| = \sqrt{b_1^2 + \cdots + b_m^2}. \tag{3}$$

Example 2. *Let* $a_n = \begin{pmatrix} \frac{n+1}{n} \\ \frac{1}{n} \end{pmatrix}$. *So, the sequence starts* $\begin{pmatrix} \frac{2}{1} \\ \frac{1}{1} \end{pmatrix}, \begin{pmatrix} \frac{3}{2} \\ \frac{1}{2} \end{pmatrix}, \begin{pmatrix} \frac{4}{3} \\ \frac{1}{3} \end{pmatrix} \ldots$. *I claim that this sequence converges to* $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$. *Note that* $||a_n - \begin{pmatrix} 1 \\ 0 \end{pmatrix}|| = ||\begin{pmatrix} \frac{1}{n} \\ \frac{1}{n} \end{pmatrix}|| = \sqrt{\frac{1}{n^2} + \frac{1}{n^2}} = \frac{\sqrt{2}}{n}$ *which is less or equal to* $\varepsilon > 0$ *if* $n \geq \frac{\sqrt{2}}{\varepsilon}$. *Hence, choosing* $N = \left[\frac{\sqrt{2}}{\varepsilon}\right] + 1$ *we have shown that* $||a_n - \begin{pmatrix} 1 \\ 0 \end{pmatrix}|| \leq \varepsilon$ *for all* $n \geq N$ *and therefore* $(a_n)_{n=1}^{\infty}$ *converges indeed to* $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$.

We can define closed sets in $\mathbb{R}^m$ in terms of sequences:

Definition 1. *A set* $S \in \mathbb{R}^m$ *is* ***closed*** *if and only if every converging sequence in* S *has its limit in* S.

Put differently, let $(s_n)_{n=1}^{\infty}$ be an arbitrary sequence in S, i.e. $s_n \in S$ for $n = 1, 2, \ldots$, and let $(s_n)_{n=1}^{\infty}$ converge to s. Then $s \in S$ if S closed. If $s \in S$ for every converging sequence (that is entirely contained in S), then S is closed.

Example 3. *Let's look at the simple case where* $m = 1$, *i.e. we have normal real sequences. The interval* $S = (1, 2]$ *(including 2 but not including 1) is* not *closed: Our*

sequence from the first example $2, \frac{3}{2}, \frac{4}{3}, \frac{5}{4}, \ldots, \frac{n+1}{n}, \ldots$ *is in* S*, i.e. each* a_n *is an element of* S*. But the limit of the sequence is 1 which is not in* S*.*

Now let's look at the set $S' = [1, 2]$*. This set is closed and we can show this by contradiction: Suppose,* S' *was not closed. Then there would be a sequence* $(a_n)_{n=1}^{\infty}$ *such that (i)* $a_n \in S'$ *for all* n*, (ii)* $(a_n)_{n=1}^{\infty}$ *converges to* a *and (iii)* $a \notin S'$*. Hence,* a *would be either strictly greater than 2 or strictly smaller than 1. Let's say* a *was strictly greater than 2 (the other case is similar). Then define* $\varepsilon = (a-2)/2$*. As* $(a_n)_{n=1}^{\infty}$ *converges to* a*, there would have to be some* a_n *such that* $|a_n - a| < \varepsilon$*. But this would imply that* $a_n > 2$ *and therefore* $a_n \notin S'$*. But* $a_n \notin S'$ *contradicts (i). Hence,* S' *is closed.*

One important concept is a *subsequence*. Intuitively, a subsequence is what is left over after you delete some terms of the original sequence. For example, take the sequence $(1/1, 1/2, 1/3, 1/4, 1/5, \ldots)$. One subsequence of this sequence would be $(1/1, 1/2, 1/4, 1/8, 1/16 \ldots)$. Another subsequence would be $(1/2, 1/4, 1/10, 1/28, \ldots)$ (in case you wondered: the rule here is $1/(3^{n-1}+1)$). For a more formal definition of a subsequence, check your math books, e.g. Simon/Blume ch. 12 (p.256).

One theorem that we need a couple of times is the Bolzano-Weierstrass theorem.[1]

Theorem 1. *Let* S *be a compact set in* $\mathbb{R}^m$ *and let* $(a_n)_{n=1}^{\infty}$ *be a sequence that is entirely contained in* S*, i.e.* $a_n \in S$ *for all* $n = 1, 2 \ldots$*. Then* $(a_n)_{n=1}^{\infty}$ *has a convergent subsequence whose limit lies in* S*.*

Instead of a full-fledged proof we just look at the idea for the simple case where $m = 1$, i.e. the sequence is a sequence of real numbers. Then our compact set S is a closed interval and for simplicity we let $S = [0, 1]$. Now we construct a subsequence $(b_n)_{n=1}^{\infty}$ of $(a_n)_{n=1}^{\infty}$ such that $(b_n)_{n=1}^{\infty}$ converges:

1. Let's split $[0, 1]$ in two subintervals $[0, 1/2]$ and $[1/2, 1]$. At least one of the two subintervals will contain an infinite number of a_n. If $[0, 1/2]$ contains an infinite number of elements of $(a_n)_{n=1}^{\infty}$, then we choose b_1 as an arbitrary a_{n_1} in the

[1] Recall: A set S in $\mathbb{R}^m$ is called *compact* if and only if it is closed and bounded. Boundedness means that there is a $K \in \mathbb{R}^m$ such that $s < K$ and $s > -K$ for all $s \in S$, i.e. the elements of S are not arbitrarily large (in absolute value).

subinterval $[0, 1/2]$. If $[0, 1/2]$ does not contain an infinite number of a_ns, we let b_1 be some arbitrary a_{n_1} in $[1/2, 1]$.

2. Now we split $[0, 1]$ into four subintervals: $[0, 1/4]$, $[1/4, 1/2]$, $[1/2, 3/4]$ and $[3/4, 1]$. We let b_2 be an arbitrary a_{n_2} that satisfies 2 conditions: (i) $n_2 > n_1$ and (ii) a_{n_2} is in the lowest of the four subintervals that contains an infinite number of a_ns.

3. Now we split $[0, 1]$ into 16 equally long closed subintervals. We let b_3 be some a_{n_3} such that: (i) $n_3 > n_2$ and (ii) a_{n_3} is in the lowest of the 16 subintervals that contains an infinite number of a_ns.

4. ...

The sequence $(b_n)_{n=1}^{\infty}$ constructed in this way converges: By the construction, any two b_n and $b_{n'}$ for $n > N$ and $n' > N$ are no more than $(1/2)^N$ apart. This means that the sequence $(b_n)_{n=1}^{\infty}$ is a so called "Cauchy sequence" and a fundamental property of the real numbers is that every Cauchy sequence converges.[2] As $[0, 1]$ is closed (and all b_n are in $[0, 1]$, the limit of this sequence will be in $[0, 1]$.

For a more detailled exposition, you can check almost any "Maths for economists" book (there are a lot of them). One (of many) that is good to read is

Carl P. Simon and Lawrence Blume: *"Mathematics for Economists",* W.W. Norton & Company Inc., 1994

where all these results (and many more) can be found in chapter 12 (and some more advanced results in chapter 29).[3]

[2]In principle, this is something I should prove but I guess that this property is not super surprising and I leave it here.

[3]In case you are very much intrigued by these sort of results and want to know more even after reading Simon/Blume, you can have a look at de la Fuente's "Mathematical Methods and Models in Economics" or Efe Ok's "Real Analysis with Economic Applications".